A Literature Review on IoT Based Applications

Miss. Priyanka S. Ghadage, Prof. S.R.Khot,
Student, Assistant Professor,
Department of Electronics and telecommunication,
DYPCET, Kolhapur, India
priyanikam1107@gmail.com , srkhot.dypcet@dypgroup.edu.in.

Abstract - **The Internet of Things (IoT) is one of the most rapidly evolving technological innovations of this century. Its key components infrastructure, architecture, and security—play a vital role in shaping the foundation of a highly digitalized future. The proliferation of connected devices, collectively referred to as the IoT, is increasing at an unprecedented rate. Consequently, network frameworks must be capable of supporting these devices by ensuring robust connectivity and delivering efficient, application-oriented services. As IoT emerges as the next significant phase in the evolution of the Internet, it becomes essential to identify its potential application domains and explore the challenges associated with its widespread adoption. Traditionally, communication has occurred between humans or between humans and devices. However, IoT envisions a future dominated by machine-to-machine (M2M) interactions, transforming the way data is exchanged and processed. The growing appeal of IoT lies in its innovative and interactive nature, which promises to revolutionize both personal and industrial environments. This paper aims to provide a comprehensive overview of the IoT landscape, including its enabling technologies, architectural frameworks, and prospective applications. Furthermore, it highlights the key challenges and critical issues that must be addressed to ensure the secure and efficient implementation of IoT systems.**

Index Terms - **Internet of Things (IoT), Network, Architecture, Security, Challenges, IoT Applications, RFID Authentication, Internet of Everything (IoE), Internet of Nano-things (IoNT) .**

I. INTRODUCTION

The Internet of Things (IoT), often referred to as the Internet of Everything (IoE) or the Industrial Internet, represents a modern technological paradigm that connects a network of intelligent and interactive machines or devices. This emerging field is becoming increasingly crucial for the advancement of future technologies and has been attracting significant attention from researchers, industries, and users worldwide (Malik et al., 2019). As defined by Nimodia and Ajankar (2022), IoT is a comprehensive strategy that integrates data from various devices across existing internet infrastructures. The primary objective of IoT is to enable seamless connectivity among digital devices, allowing communication between objects, systems, and individuals across multiple networks and services—anytime and anywhere (Hussein, 2019). It envisions an intelligent and interconnected world that supports dynamic interaction between humans and their environment. Mobile phones play a crucial role in authenticating and interacting with connected devices, employing the concept of object hyperlinks, which extend internet connectivity to physical objects and real-world locations (Tyagi et al., 2014).

The Internet has thus evolved beyond being a mere network of computers into a network of diverse devices, forming what Hussein (2019) describes as a "network of networks." Although large-scale commercial adoption is still evolving, IoT offers immense opportunities for business innovation, academic research, and technological advancement. Consequently, the study of IoT focuses on identifying potential application domains and research challenges associated with its implementation (Hussein, 2019).With its growing importance in everyday life, IoT continues to integrate intelligent systems, smart devices, and adaptive frameworks to enhance human experience (Kumar et al., 2019). From an architectural standpoint, IoT systems connect numerous sensors and intelligent devices to local or cloud-based controllers, enabling real-time data collection and automated decision-making (Yaici et al., 2021). Typically, IoT architecture comprises three primary layers:

Perception or Sensor Layer – includes physical devices with unique identities capable of sensing and transmitting data without human intervention.

Network or Control Layer – facilitates communication and data processing.

Application Layer – delivers user-oriented services and interfaces.

However, several challenges persist, particularly regarding the global distribution and accessibility of IoT devices. Ensuring seamless connectivity across dispersed devices requires robust network infrastructure and significant investment from service providers (Alenezi et al., 2021). Furthermore, security remains a major concern, as IoT systems must safeguard their hardware, networks, and transmitted data from potential threats (Mohamad Noor & Hassan, 2019).

This paper aims to explore the core concepts, infrastructure, and architecture of the Internet of Things while highlighting its security concerns, research challenges, and future opportunities that promise to make human life more efficient and connected than ever before.

II. LITERATURE REVIEW

The current Internet architecture, which uses the TCP/IP protocol, was adopted in 1980, but by 2030, more than 25.4 billion people are anticipated to be connected, which is a tremendous figure (Nimodia & Ajankar, 2022). It requires a new open architecture to address numerous security and quality of service (QoS) challenges as well as support current network applications using open

protocols because the IoT cannot handle it (M. Zhang et al., 2012). Without adequate privacy protection, it is unlikely that many people will adopt IoT (Suo et al., 2012). Therefore, data security and user privacy are critical IoT challenges. Several layered security architectures have been proposed for additional development of the IoT. (Chen, 2013) described an architecture with three significant layers of IoT, and (Suo et al., 2012) described an architecture with four important layers. (Wu et al., 2010) suggested a five-tier architecture that leverages the best Internet and communications management network architectures based on the TCP / IP and TMN models. Similarly, a 6-tier architecture based on a hierarchical network structure has been proposed (Suo et al., 2012). Consequently, it is typically separated into six layers. The following describes each of the five IoT levels.

Coding layer-The IoT's coding layer, which identifies the target object, is its core. At this layer, each object is given a distinct ID, making it simple to identify between things (M. Zhang et al., 2012).

Perceptual layer- Each object has a physical meaning gives by the layer. It is made up of several data sensors that may measure an object's temperature, humidity, velocity, position, and other properties (Bandyopadhyay & Sen, 2011). This layer gathers important data from the sensing devices about an object and transforms it into a digital signal. The network layer receives the digital signal to take further action.

Network layer- This layer's function is to gather useful data from the perception layer in the form of digital signals and transmit it over transmission mediums such WiFi, Bluetooth, WiMAX, ZigBee, GSM, and 3G to the middleware layer's processing system. It employs MQTT, DDS, IPv4, and IPv6 (Y. Zhang, 2011).

Middleware layer- The data sent by the sensor device is handled by this layer (Shen & Liu, 2011). It includes innovations like cloud computing and ubiquitous computing that let you store all the data you require and instantly access the database. Information is processed using some intelligent processing tools, and the results of the processing are totally automated activities.

Application layer -Based on processed data, this layer enables IoT applications for various industries. As applications support IoT development, this layer is advantageous for the broad expansion of IoT networks (Wu et al., 2010).

IoT Technologies

To enable seamless identification and communication among connected devices, it is essential to develop a set of advanced and efficient technologies within ubiquitous computing systems. These systems should be capable of uniquely identifying digital objects, interacting intelligently with other entities, and collecting data to perform automated operations (Khoo, 2011). The extensive growth and functionality of the Internet of Things (IoT) largely depend on the integration of these interconnected technologies discussed in this section.

Radio Frequency Identification (RFID)

RFID is an important tech that allows things to be identified. Due to its reduced size and cost can be embedded in anything (Chen, 2013). Depending on the application, RFID is a transmitter microchip that resembles a sticker and can be either active or passive (Guo et al., 2011). RFID systems include RFID tags and readers that convey information about an object, such as its ID or position, when the right signal is generated (Zhang & Zhu, 2011). Using radio frequencies, the reader transmits the object-related data signal that was emitted, which is then sent on to the processor for analysis. The creation of a universal computing system where digital things may be precisely recognized and have the capacity to think and engage in interactions with other objects to gather facts depending on whether automated operations are conducted, a mix of modern, efficient technology is required, which can only be accomplished by integrating several technologies that enable object identification and communication (Khoo, 2011). In this part, essential technologies are covered that can aid in the widespread growth of IoT. RFID frequencies are divided into four different frequency ranges based on the type of application (Guo et al., 2011).

Wireless Sensor Network (WSN)

A Wireless Sensor Network (WSN**)** is a crucial enabling technology in the Internet of Things (IoT) that facilitates the sensing, collection, and transmission of data from the physical environment to computational systems. A WSN consists of a large number of small, low-power, and spatially distributed devices known as sensor nodes, which are capable of sensing environmental parameters, performing local processing, and communicating wirelessly with other nodes or a central base station. WSN is a bi-directional wirelessly related community of sensors in a multi-hop fashion, constructed from numerous nodes scattered in a sensor subject every about one or multiple sensors which could gather object-unique statistics which include temperature, humidity, velocity after which by skipping directly to the processing equipment (Atzori et al., 2010). Sensing nodes communicate using multi-hop. Each sensor is a transceiver equipped with an antenna, a microcontroller, and an interface circuit that serve as the sensors' communication, actuation, and sensing units, respectively. The power source for each sensor may be a battery or any power-harvesting technology (Tapia et al., 2007). However (Shen & Liu, 2011) has suggested a new unit, the Memory Unit that may also be a component of the sensing node, for storing the statistics. While blended collectively, the Wireless Sensors Network era and RFID era open up opportunities for even extra clever devices to some of the answers proposed (Atzori et al., 2010).

Cloud Computing

Cloud computing is a fundamental technology that supports the large-scale data storage, processing, and analytics requirements of the Internet of Things (IoT). It provides a virtualized and scalable computing environment that enables users to access computing resources—such as servers, storage, databases, and applications—over the Internet on a pay-per-use basis.

At its core, cloud computing allows organizations and IoT systems to offload data-intensive operations to remote servers, thereby reducing local processing and storage constraints. This capability is crucial for IoT, where a massive volume of sensor data is generated continuously and needs to be processed efficiently in real time. With millions of gadgets anticipated by 2020 (Kaur, 2020), the lone technology that appears to efficiently evaluate and preserve all data is the cloud. It is an advanced computer system that integrates many servers on a cloud platform to enable the mutual sharing of resources accessible anytime, anywhere (Kaur, 2020).

The crucial component of the IoT is cloud computing, consolidating servers and handling increased computing power, analyzing helpful information obtained from sensors, and providing superior storage capacity (Rekha et al., 2021).

Network Technology:

Network technology serves as the backbone of communication in the Internet of Things (IoT) ecosystem. It enables seamless data transmission, device connectivity, and real-time interaction among millions of IoT devices distributed across various locations. k. Networks for long-distance transmission often employ 3G, 4G, etc. However, as you are aware, mobile data traffic is predictable because it is only required for everyday operations like placing calls and sending texts. 5th generation wireless systems must be extremely quick, extremely efficient, unexpected, and able to supply far more bandwidth in the present era of ubiquitous computing (Vermesan & Friess, 2014).

Nanotechnology:

This technology provides a more compact and advanced evolution of interconnected systems. By enabling the creation of devices at the nano meter scale, it significantly reduces system power consumption while maintaining functionality similar to conventional sensors and actuators. These nano devices are fabricated using nanoscale components, and their integration gives rise to a novel network architecture known as the Internet of Nano Things (IoNT) (Atlam et al., 2018).

IoT Applications Given that Internet of Things affects virtually every aspect of people's, organizations', and civilizations' daily lives, and has a vast range of possible uses. According to (Patel et al., 2016) IoT applications span a wide range of industries, including manufacturing or industrial sectors, healthcare sectors, agriculture, smart cities, security, and emergencies.

The Internet of Things (IoT) would enhance the overall infrastructure and cities' smartness (Zanjal & Talmale, 2016). Areas of IoT applications in the creation of smart cities include sophisticated transportation infrastructure (Jain, 2018), innovative construction, traffic congestion and waste management (Bellini et al., 2022), bright lighting, smart parking, and city maps. Monitoring municipal parking spots, keeping tabs on building vibration and physical condition, placing noise-monitoring equipment in sensitive places, and keeping an eye on pedestrian and vehicle activity are just a few features that may be added. Traffic congestion in smart cities can be tracked, managed, and reduced with IoT powered by artificial intelligence (AI) (Zanjal & Talmale, 2016). In addition, IoT technology can be utilized to implement smart, weather-adaptive street lighting systems and to monitor waste collection schedules for efficient debris management. Intelligent roadways equipped with IoT sensors can transmit crucial data and real-time alerts. These systems enable adaptive responses to environmental conditions **and** unexpected events**,** such as traffic congestion or accidents, by providing timely information and alternative route suggestions.

Healthcare:

The majority of healthcare systems worldwide are ineffective, cumbersome, and invariably prone to mistakes. The healthcare industry depends on a wide range of processes and equipment, so technology automation and expansion are readily modifiable. The healthcare industry will undergo a massive transformation thanks to new technologies that make a variety of tasks more manageable, such as exchanging reports with numerous individuals and places, storing records, and distributing medications (Mano et al., 2016). IoT applications for health may provide many advantages that can be broadly grouped into the following areas: patient, staff, and asset monitoring, personal identity and authentication, automated data collecting and processing, and tracking. Applications. In this domain include monitoring patient adherence to medication, telemedicine solutions, and real-time health alert systems. IoT-enabled sensors can collect and transmit data during and after patient monitoring, making them applicable for both inpatient and outpatient care. They can also be integrated into dental Bluetooth devices and smart toothbrushes to support continuous health tracking.

In this context, technologies such as RFID, Bluetooth, and Wi-Fi play a crucial role in enabling seamless data communication and integration. These technologies significantly enhance methods for measuring and monitoring vital health parameters, including blood pressure, body temperature, heart rate, blood glucose, and cholesterol levels. Furthermore, the emergence of the Internet of Nano Things (IoNT) has expanded the capabilities of the Internet of Things (IoT) and Internet of Everything (IoE), enabling even more precise and intelligent healthcare applications (Miraz et al., 2015).

Smart Agriculture and Water Management

According to Badran and Kashmoola (2020)**,** the integration of Internet of Things (IoT) technologies in the agricultural sector can significantly enhance productivity, sustainability, and resource utilization. In applications such as vineyard monitoring, IoT systems can measure soil moisture and stem diameter, enabling precise irrigation control and crop growth analysis. Through the deployment of IoT sensors and connected systems, farmers can effectively manage microclimate conditions, optimize the yield and quality of fruits and vegetables, and even monitor the nutritional content—including vitamin levels—of agricultural produce.

By continuously analyzing weather patterns, IoT platforms can regulate parameters such as temperature and humidity**,** thereby reducing the risk of fungal infections and microbial contamination. Furthermore, IoT-based predictive models can help forecast extreme climatic events such as frost, droughts, wind fluctuations, rainfall, and snow, enabling proactive agricultural management.

In livestock management, IoT facilitates animal identification**,** health monitoring, and detection of hazardous gases produced from animal waste. Such implementations improve animal welfare, increase survival rates, and optimize breeding and growth conditions.

In the context of water management, IoT plays a critical role in assessing water quality and **s**uitability for both drinking and agricultural purposes. It also enables the monitoring of pressure variations in pipelines and storage tanks, as well as the detection of liquid levels in dams, rivers, and reservoirs to prevent shortages or flooding. These applications are primarily supported by Wireless Sensor Networks (WSNs) that ensure seamless data transmission and environmental sensing.

Smart Living

In this domain, Internet of Things (IoT) technology enables the remote control and automation of household appliances, allowing users to operate devices from distant locations to prevent accidents and conserve energy (Miraz et al., 2018). Smart home appliances, such as refrigerators equipped with Liquid Crystal Display (LCD) panels, can provide detailed information about stored items—indicating which products are available, nearing expiration, or need replenishment. This data can be synchronized with a mobile application, enabling users to access inventory information remotely and make timely purchasing decisions even while away from home. Similarly, smart washing machines allow users to monitor and manage laundry cycles remotely, while various connected kitchen appliances can be operated through smartphones—for instance, adjusting oven temperatures or activating self-cleaning features. These innovations enhance convenience, efficiency, and time management in daily household activities. IoT plays a vital role in enhancing home security systems. Through IoT-enabled surveillance cameras, users can remotely monitor their homes, detect unauthorized entry through windows or doors, and receive instant alerts. Integrated alarm systems further help deter potential intrusions, providing a safer and smarter living environment (Miraz et al., 2015).

IV.CHALLENGES AND ISSUES WITH IOT:

For each of the potential IoT applications discussed above, the effectiveness and functionality of these systems depend greatly on their practical feasibility across various domains. Similar to other emerging technologies, the Internet of Things (IoT) faces several challenges and **i**mplication**s** that must be addressed before achieving large-scale adoption. Although current IoT technologies have advanced considerably in recent years, numerous unresolved issues remain, opening new avenues for future research.The concept of IoT has evolved from a convergence of diverse technologies designed to collect, process, generate, exchange, transmit, manage, and store data. This integration, however, introduces multiple technical and research challenges related to interoperability, scalability, security, and standardization. Consequently, these challenges have expanded the IoT research landscape, spanning multiple disciplines and creating opportunities for continued exploration and innovation (Rehman et al., 2018).

Privacy and Security

Internet of Things (IoT) features need to be completely taken into account as Internet usage increases and the IoT emerges as a crucial part of the Internet's future. Many modern IoT devices have weaknesses, which researchers are aware of. Additionally, because the Internet of Things (IoT) is based on already-existing wireless sensor networks (WSNs), it shares WSNs' privacy and security flaws (Alansari et al., 2018). Many assaults often take use of weaknesses in certain devices to obtain access to the system and, as a result, being vulnerable to secure devices (Abu Daia et al., 2018; Chaudhry et al., 2018). In addition to the safety and protection mechanisms within IoT systems, several other critical aspects—such as communication confidentiality, system reliability, connectivity trustworthiness, message integrity, and enhanced security requirements—must also be addressed. It is equally important to prevent unauthorized communication or data exchange between untrusted entities. For instance, smart devices must ensure that sensitive information is safeguarded against access or misuse by competitors, thereby preventing potential data breaches or malicious exploitation in commercial and transactional contexts.

For each of the potential IoT applications discussed above, the effectiveness and functionality of these systems depend greatly on their practical feasibility across various domains. Similar to other emerging technologies, the Internet of Things (IoT) faces several challenges and implications that must be addressed before achieving large-scale adoption. Although current IoT technologies have advanced considerably in recent years, numerous unresolved issues remain, opening new avenues for future research.The concept of IoT has evolved from a convergence of diverse technologies designed to collect, process, generate, exchange, transmit, manage, and store data. This integration, however, introduces multiple technical and research challenges related to interoperability, scalability, security, and standardization. Consequently, these challenges have expanded the IoT research landscape, spanning multiple disciplines and creating opportunities for continued exploration and innovation (Rehman et al., 2018).

Interoperability and Standardization:
Traditionally, interoperability is the fundamental core of the Internet. The essential requirement for an Internet connection requires that "connected" systems be able to "speak similar languages" concerning cryptography and protocols. It's valued, and it's still essential. Currently, different things are used in various industries. The absence of universal standards limits seamless interaction and integration between heterogeneous systems.

Scalability:
As the number of connected devices grows exponentially, managing data traffic, bandwidth, and network performance becomes increasingly complex.

Data Management and Storage:
IoT generates massive volumes of real-time data. Efficient methods for data collection, processing, analysis, and storage are required to ensure system responsiveness and reliability.

Energy Efficiency:
Many IoT devices operate on limited power sources such as batteries. Designing low-power communication protocols and energy-efficient hardware is essential to extend device lifespan.

Network Reliability and Connectivity:
Maintaining consistent connectivity, especially in remote or mobile environments, is challenging. Factors such as latency, signal interference, and packet loss can degrade system performance.

Cost and Infrastructure:
Deploying and maintaining large-scale IoT networks require significant investment in hardware, cloud infrastructure, and skilled personnel.

Legal and Ethical Issues:
The use of IoT data raises questions about data ownership, surveillance, and ethical use**.** Regulatory frameworks are still evolving to ensure responsible data governance.

V.CONCLUSION

The Internet of Things (IoT**)** can be defined as a dynamic and adaptive network of interconnected systems, necessitating innovative approaches in software engineering, systems design, project management, and other related disciplines to support its continuous evolution and effective management in the coming years. A five-layer IoT architecture has been proposed to structure its operation and development.

Organizations aiming to harness the full potential of IoT must first establish a well-defined strategic *framework* to address existing and emerging security vulnerabilities. Developing an architecture-focused strategy not only enhances successful implementation but also minimizes potential security risks**,** thereby improving the reliability and trustworthiness of IoT applications.

One of the primary challenges in IoT deployment is achieving a balance between security and usability, ensuring that robust protection measures do not compromise device or network performance. Future research is expected to further strengthen the IoT ecosystem through the integration of advanced technologies such as statistical fingerprinting and block chain, which can enhance authentication, transparency, and data integrity.

Ultimately, this rapidly evolving paradigm will profoundly influence every aspect of human life—from **s**mar**t** homes and healthcare **to** environmental monitoring and industrial automation—by embedding intelligence and connectivity into everyday objects. As a result, IoT stands as a transformative force that serves individuals, communities, organizations, and society as a whole**,** shaping a smarter and more connected future.

VI. ACKNOWLEDGMENT

It gives me immense pleasure to acknowledge and thanks to many people who contributed in various ways for the successful completion of this research work. I take this opportunity with pleasure to express my deep sense of gratitude to my guide, **Prof. S. R. Khot** for his excellent guidance, encouragement, support and useful suggestions throughout the period of making this review paper.

REFERENCES

[1] Nimodiya, A. R., & Ajankar, S. S. (2022). A Review on Internet of Things. International Journal of Advanced Research in Science, Communication and Technology, 113(1), 135–144. https://doi.org/10.48175/ijarsct2251.

[2] Bellini, P., Nesi, P., & Pantaleo, G. (2022). IoT-Enabled Smart Cities: A Review of Concepts, Frameworks and Key Technologies. Applied Sciences (Switzerland), 12(3). https://doi.org/10.3390/app12031607.

[3] Yaïci, W., Krishnamurthy, K., Entchev, E., & Longo, M. (2021). Recent advances in Internet of things (IoT) infrastructures for building energy systems: A review. In Sensors (Vol. 21, Issue 6). https://doi.org/10.3390/s21062152.

[4] Kaur, C. (2020). The Cloud Computing and Internet of Things (IoT). International Journal of Scientific Research in Science, Engineering and Technology, January, 19–22. https://doi.org/10.32628/ijsrset196657.

[5] Nikhat Akhtar, & Yusuf Perwej. (2020). The Internet of nano things (IoNT) existing state and future Prospects. GSC Advanced Research and Reviews, 5(2), 131–150. https://doi.org/10.30574/gscarr.2020.5.2.0110.

[6] Malik, A., Magar, A. T., Verma, H., Singh, M., & Sagar, P. (2019). A detailed study of the internet of things (Iot). International Journal of Scientific and Technology Research, 8(12), 2989–2994.

[7] Hussein, A. R. H. (2019). Internet of Things (IOT): Research challenges and future applications. International Journal of Advanced Computer Science and Applications, 10(6), 77–82. https://doi.org/10.14569/ijacsa.2019.01

[8] Kumar, S., Tiwari, P., & Zymbler, M. (2019). Internet of Things is a revolutionary approach for future technology enhancement: a review. Journal of Big Data, 6(1). https://doi.org/10.1186/s40537-019-0268-2.

[9] Miraz, M. H. (2019). Blockchain of Things (BCoT): The Fusion of Blockchain and IoT Technologies. In SSRN Electronic Journal (Issue September). https://doi.org/10.2139/ssrn.3464085.

[10] Mohamad Noor, M. binti, & Hassan, W. H. (2019). Current research on the Internet of Things (IoT) security: A survey. Computer Networks, 148, 283–294. https://doi.org/10.1016/j.comnet.2018.11.025.

[11] Rehman, H. U., Asif, M., & Ahmad, M. (2018). Future applications and research challenges of IoT. 2017 International Conference on Information and Communication Technologies, ICICT 2017, 2017-December (December 2017), 68–74. https://doi.org/10.1109/ICICT

[12] Miraz, M. H., Ali, M., Excell, P. S., & Picking, R. (2018). Internet of Nano-Things, things and everything: Future growth trends. Future Internet, 10(8). https://doi.org/10.3390/fi10080068.

[13] Jain, R. (2018). A congestion control system based on vanet for small length roads. Annals of Emerging Technologies in Computing, 2(1), 17–21. https://doi.org/10.33166/AETiC.2018.01.003. 23.

[14] Mano, L. Y., Faiçal, B. S., Nakamura, L. H. V., Gomes, P. H., Libralon, G. L., Meneguete, R. I., Filho, G. P. R., Giancristofaro, G. T., Pessin, G., Krishnamachari, B., & Ueyama, J. (2016). Exploiting IoT technologies for enhancing Health Smart Homes through

patient identification and emotion recognition. Computer Communications, 89–90, 178–190. https://doi.org/10.1016/j.comcom.2016.03.010.

[15] Patel, K. K., Patel, S. M., & Scholar, P. G. (2016). Internet of Things-IoT: Definition, Characteristics, Architecture, Enabling Technologies, Application & Future Challenges. International Journal of Engineering Science and Computing, 6(5), 1–10. https://doi.org/10.4010/2016.1482

[16] Zanjal, S. V., & Talmale, G. R. (2016). Medicine Reminder and Monitoring System for Secure Health Using IoT. Physics Procedia, 78(August), 471–476. https://doi.org/10.1016/j.procs.2016.02.090

[17] Miraz, M. H., Ali, M., Excell, P. S., & Picking, R. (2015). A review on Internet of Things (IoT), Internet of Everything (IoE) and Internet of Nano Things (IoNT). 2015 Internet Technologies and Applications, ITA 2015 - Proceedings of the 6th International Conference, September, 219–224. https://doi.org/10.1109/ITechA.2015.7317398.

[18] Vermesan, O., & Friess, P. (2014). Internet of things applications: From research and innovation to market deployment. Internet of Things Applications: From Research and Innovation to Market Deployment, June, 1– 371.

[19] Tyagi, S., Darwish, A., & Khan, M. Y. (2014). Managing Computing Infrastructure for IoT Data. Advances in Internet of Things, 04(03), 29–35. https://doi.org/10.4236/ait.2014.43005

[20] Zhang, M., Sun, F., & Cheng, X. (2012). Architecture of Internet of Things and its key technology integration based-on RFID. Proceedings - 2012 5th International Symposium on Computational Intelligence and Design, ISCID 2012, 1, 294–297. https://doi.org/10.1109/ISCID.2012.81.

[21] Zhang, Y. (2011). Technology framework of the Internet of Things and its application. 2011 International Conference on Electrical and Control Engineering, ICECE 2011 - Proceedings, 4109–4112. https://doi.org/10.1109/ICECENG.2011.6057290.

[22] Bandyopadhyay, D., & Sen, J. (2011). Internet of things: Applications and challenges in technology and standardization. Wireless Personal Communications, 58(1), 49–69. https://doi.org/10.1007/s11277-011- 0288-5.

[23] Zhang, H., & Zhu, L. (2011). Internet of Things: Key technology, architecture and challenging problems. Proceedings - 2011 IEEE International Conference on Computer Science and Automation Engineering, CSAE 2011, 4, 507–512. https://doi.org/10.1109/CSAE.2011.5952899.

[24] Khoo, B. (2011). RFID As an enabler of the Internet of things: Issues of security and privacy. Proceedings - 2011 IEEE International Conferences on Internet of Things and Cyber, Physical and Social Computing, IThings/CPSCom 2011, October 2011, 709–712. https://doi.org/10.1109/iThings/CPSCom.2011.83.

[27] Wu, M., Lu, T. J., Ling, F. Y., Sun, J., & Du, H. Y. (2010). Research on the architecture of Internet of Things. ICACTE 2010 - 2010 3rd International Conference on Advanced Computer Theory and Engineering, Proceedings, 5(September 2010). https://doi.org/10.1109/ICACTE.2010.5579493.

[28] Yun, M., & Yuxin, B. (2010). Research on the architecture and key technology of Internet of Things (IoT) applied on smart grid. 2010 International Conference on Advances in Energy Engineering, ICAEE 2010, 69– 72. https://doi.org/10.1109/ICAEE.2010.5557611.